

# STRUCTURE, INVARIANTS AND OBSTRUCTIONS IN COLLATZ DYNAMICS: A 2-ADIC OBSTRUCTION TO MODULAR LYAPUNOV FUNCTIONS AND EXACT WASSERSTEIN CONTRACTION COEFFICIENTS

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ABSTRACT. We study the accelerated Collatz map  $T(n) = n/2$  ( $n$  even),  $T(n) = (3n + 1)/2$  ( $n$  odd), and its affine siblings  $3n + d$ . Our main result is a nonexistence theorem: for every  $j \geq 1$  there is no function  $V(n) = \log_2 n + w(n \bmod 2^j)$ , with  $w : \mathbb{Z}/2^j\mathbb{Z} \rightarrow \mathbb{R}$  arbitrary, that decreases strictly along the odd orbits of  $T$  in  $\mathbb{Z}_+$ . The obstruction is structural: the 2-adic fixed point  $-1 \in \mathbb{Z}_2$  projects, in every modulus  $2^j$ , onto a residue class carrying a self-loop of logarithmic gain  $\log_2 3 - 1 > 0$  that no modular potential can compensate, and the mechanism is robust to finite sets of exceptions, to non-strict decrease, and to conditions imposed on blocks of  $k$  iterations. We complement this with two contraction theorems for the transfer structure of the map: the Syracuse Markov chain on  $\mathbb{Z}_3$  has Dobrushin–Wasserstein coefficient at most  $1/3$  (each inverse branch scales the 3-adic metric by exactly  $1/3$ ), yielding a unique invariant measure and geometric  $W_1$ -equidistribution; and the dual chain on  $\mathbb{Z}_2$  has coefficient exactly  $1/2$  and averages every level- $k$  cylinder function to its exact Haar mean in  $k$  steps. All computational claims are produced by exact algorithms — rational arithmetic, arbitrary-precision integers, and integer certificates for the continued fraction of  $\log_2 3$  — implemented in a small, dependency-free Python library; every number quoted in the paper is regenerated by a single reproduction script. The computations instantiate, but never replace, the proofs: numerical sweeps are labelled as verifications, with limits stated explicitly.

## 1. INTRODUCTION

Let  $T : \mathbb{Z} \rightarrow \mathbb{Z}$  be the accelerated Collatz map,

$$T(n) = \begin{cases} n/2, & n \text{ even,} \\ (3n + 1)/2, & n \text{ odd.} \end{cases}$$

The Collatz conjecture asserts that every  $n \geq 1$  reaches the cycle  $\{1, 2\}$  under iteration of  $T$ . The problem has an extensive literature, surveyed by Lagarias [6, 7]; the strongest partial results include Terras’s theorem that almost every  $n$  has finite stopping time [10], the density bounds of Krasikov

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and Lagarias [5], lower bounds on the length of hypothetical nontrivial cycles by Eliahou [2] and Simons–de Weger [8], Tao’s theorem that almost all orbits attain almost bounded values [9], and the computational verification of convergence for all  $n \leq 2^{68}$  by Barina [1]. Functional-analytic and Markov-chain approaches to the  $3n+1$  map, close in spirit to our Section 4, are developed systematically by Wirsching [12].

This paper has two aims. The first is to prove a nonexistence theorem for a natural class of candidate Lyapunov functions (Section 3) and two exact contraction theorems for the transfer structure of the map on the 2-adic and 3-adic completions (Section 4). The second is methodological: we describe a small computational laboratory, built exclusively on exact arithmetic, whose outputs instantiate the theory — enumerating cycles as solutions of a Diophantine equation, certifying continued-fraction data by integer comparisons, and computing Wasserstein contraction coefficients as exact rationals (Sections 5–7).

### 1.1. Main contributions.

- (1) **A modular obstruction theorem** (Theorem 3.2). For every  $j \geq 1$  there is no strictly decreasing function of the form  $V(n) = \log_2 n + w(n \bmod 2^j)$  along odd orbits in  $\mathbb{Z}_+$ . The proof isolates a single structural cause: the fixed point  $T(-1) = -1$  in  $\mathbb{Z}_2$  projects onto the residue  $-1 \bmod 2^j$ , and the family  $n_m = 2^{j+1}m - 1$  makes the modular potential cancel exactly while the logarithmic term grows. Four robustness corollaries (Corollary 3.4) show that the usual relaxations — finitely many exceptions, non-strict decrease, decrease along blocks of  $k$  iterations, restricted domains — do not evade the obstruction.
- (2) **Exact contraction coefficients** (Theorems 4.3 and 4.5). The Syracuse Markov chain on  $\mathbb{Z}_3$  is a uniformly contractive iterated function system: every inverse branch scales the 3-adic metric by exactly  $1/3$ , so the Dobrushin–Wasserstein coefficient of the chain, and of each of its finite projections modulo  $3^k$ , is at most  $1/3$ . Consequently the chain has a unique invariant measure (Tao’s Syracuse measure [9]) and equidistributes at rate  $3^{-n}$  in  $W_1$ . Dually, the backward chain on  $\mathbb{Z}_2$  has coefficient exactly  $1/2$ , preserves Haar measure, and maps every level- $k$  cylinder function to its exact Haar average in  $k$  steps.
- (3) **An exact computational laboratory** (Sections 5–7). Cycle enumeration via the closed form  $n = b(p)/(2^L - 3^k)$ ; cycle exclusion via certified convergents of  $\log_2 3$ , reproducing the method of Eliahou and Simons–de Weger [2, 8]; identification of the obstruction of Theorem 3.2 at every finite level  $j$  by Karp’s maximum-mean-cycle algorithm [3]; and exact rational evaluation of the Wasserstein coefficients  $\tau_k$  of Section 4.

**1.2. Theorems versus verifications.** We maintain a strict separation between the two kinds of statement in this paper.

- *Theorems* (Sections 3–5) are proved analytically; none of their proofs depends on a computation.
- *Verifications* (Section 7) are outputs of the algorithms of Section 5, reported with their exact sweep limits. They serve two purposes: validating the implementations against known ground truth (the negative cycles of  $3n + 1$ , the nontrivial cycles of  $3n - 1$  and  $3n + 5$ ), and instantiating the theorems at finite levels (the Karp analysis locating  $-1 \bmod 2^j$ , the coefficients  $\tau_k$  increasing towards the bound  $1/3$ ).

In particular, our convergence sieve ( $n \leq 2 \cdot 10^5$ ) makes no claim of novelty — the published record is  $2^{68}$  [1] — and is used only as a self-contained hypothesis for the exclusion bounds of Section 7.3, which we then also evaluate at the literature’s limit.

**1.3. Reproducibility.** All algorithms are implemented in a pure-Python package `collatz` (no third-party dependencies) accompanying this paper, with a test suite and a single script, `reproduce_paper_results.py`, that regenerates and checks every number quoted here (its sections R1–R8 are cross-referenced in Section 7). The arithmetic model, and why no floating-point issue can affect any certified claim, is specified in Section 6.

## 2. NOTATION AND PRELIMINARIES

$\mathbb{Z}_+$  denotes the positive integers. For a prime  $p$ ,  $\mathbb{Z}_p$  is the ring of  $p$ -adic integers,  $v_p$  the  $p$ -adic valuation and  $|x|_p = p^{-v_p(x)}$  the  $p$ -adic absolute value;  $(\mathbb{Z}_p, |\cdot|_p)$  is a compact ultrametric space.  $\nu_2(n)$  is the 2-adic valuation of an integer  $n$ . For  $x \in \mathbb{R}$ ,  $\|x\|$  denotes the distance from  $x$  to the nearest integer. Throughout,  $\alpha = \log_2 3$ .

**Definition 2.1** (Collatz maps). For an integer parameter  $d$  odd, the *accelerated map* of the  $3n + d$  system is

$$T_d(n) = \begin{cases} n/2, & n \text{ even,} \\ (3n + d)/2, & n \text{ odd,} \end{cases}$$

and we write  $T = T_1$ . The *Syracuse map*  $S$  acts on odd integers by  $S(n) = (3n + 1)/2^{\nu_2(3n+1)}$ . The *parity vector* of length  $k$  of  $n$  is  $(n \bmod 2, T(n) \bmod 2, \dots, T^{k-1}(n) \bmod 2) \in \{0, 1\}^k$ .

We will use Terras’s structural theorem: the length- $k$  parity vector, as a function of  $n \bmod 2^k$ , is a bijection of  $\mathbb{Z}/2^k\mathbb{Z}$  onto  $\{0, 1\}^k$ , and  $T$  extends to  $\mathbb{Z}_2$  where it is topologically conjugate to the one-sided shift [10, 6].

**Definition 2.2** (Wasserstein distance and contraction coefficient). Let  $(X, d)$  be a compact metric space and  $\mu, \nu$  Borel probability measures on

$X$ . The Kantorovich–Rubinstein distance is

$$W_1(\mu, \nu) = \inf_{\gamma \in \Gamma(\mu, \nu)} \int_{X \times X} d(x, y) d\gamma(x, y),$$

the infimum over couplings of  $\mu$  and  $\nu$  [11, Ch. 6]. For a Markov kernel  $P$  on  $X$ , the *Dobrushin–Wasserstein contraction coefficient* is

$$\tau(P) = \sup_{x \neq y} \frac{W_1(P(x, \cdot), P(y, \cdot))}{d(x, y)}.$$

The standard consequences we use are collected in the next lemma.

**Lemma 2.3.** *Let  $P$  be a Markov kernel on a compact metric space  $(X, d)$  with  $\tau = \tau(P) < 1$ . Then:*

- (1)  $W_1(\mu P, \nu P) \leq \tau W_1(\mu, \nu)$  for all Borel probability measures  $\mu, \nu$ ;
- (2)  $P$  has a unique invariant measure  $\pi$ , and  $W_1(\mu P^n, \pi) \leq \tau^n W_1(\mu, \pi)$  for every  $\mu$ ;
- (3) the Koopman operator  $Uf(x) = \int f dP(x, \cdot)$  satisfies  $\text{Lip}(Uf) \leq \tau \text{Lip}(f)$ ; hence on the quotient of the Lipschitz space by the constants, the spectral radius of  $U$  is at most  $\tau$ , i.e.  $\text{spec}(U|_{\text{Lip}}) \subseteq \{1\} \cup \{z : |z| \leq \tau\}$ .

*Proof.* (1) Given a coupling  $\gamma$  of  $(\mu, \nu)$  and, for each pair  $(x, y)$ , a coupling  $\gamma_{x,y}$  of  $(P(x, \cdot), P(y, \cdot))$  with cost at most  $\tau d(x, y) + \varepsilon$ , the mixture  $\int \gamma_{x,y} d\gamma$  couples  $\mu P$  and  $\nu P$  with cost at most  $\tau \int d d\gamma + \varepsilon$ ; take infima. (2)  $X$  compact implies the space of Borel probability measures with  $W_1$  is a complete metric space [11, Thm. 6.18]; by (1) the map  $\mu \mapsto \mu P$  is a  $\tau$ -contraction, so the Banach fixed point theorem applies. (3) For  $f$  Lipschitz and  $x \neq y$ , Kantorovich–Rubinstein duality [11, Ch. 5] gives  $|Uf(x) - Uf(y)| \leq \text{Lip}(f) W_1(P(x, \cdot), P(y, \cdot)) \leq \text{Lip}(f) \tau d(x, y)$ . Iterating,  $\text{Lip}(U^n f) \leq \tau^n \text{Lip}(f)$ , so on  $\text{Lip}/\text{constants}$  the operator norm of  $U^n$  is at most  $\tau^n$ .  $\square$

### 3. THE MODULAR OBSTRUCTION TO LYAPUNOV FUNCTIONS

A natural strategy for proving global convergence is to exhibit a Lyapunov function combining the macroscopic logarithmic trend of the orbit with periodic local corrections.

**Definition 3.1** (Modular Lyapunov function). Fix  $j \geq 1$ . A *modular Lyapunov function of level  $j$*  is a function

$$V(n) = \log_2 n + w(n \bmod 2^j), \quad w : \mathbb{Z}/2^j\mathbb{Z} \rightarrow \mathbb{R} \text{ arbitrary,}$$

such that  $V(T(n)) < V(n)$  for every odd integer  $n > 0$ .

Since  $w$  has finite domain, it is bounded, and on the integers it acts as a periodic potential of period  $2^j$ . If such a  $V$  existed, no orbit could diverge (the  $\log_2$  term would be forced below any bound along the infinitely many odd steps of a divergent orbit while even steps also decrease  $V$ , since

$V(n/2) = V(n) - 1 + w(\frac{n}{2} \bmod 2^j) - w(n \bmod 2^j)$  is controlled by the same potential — but we will not need this motivation). Our main result is that the class is empty.

**Theorem 3.2** (Modular obstruction). *For every  $j \geq 1$ , no modular Lyapunov function of level  $j$  exists. More precisely, the projection of the 2-adic fixed point  $-1 \in \mathbb{Z}_2$  onto  $\mathbb{Z}/2^j\mathbb{Z}$  induces, on the residue class  $-1 \bmod 2^j$ , a growth loop of logarithmic gain  $\log_2 3 - 1 > 0$  that no potential  $w$  can compensate.*

*Proof.* Suppose, for contradiction, that there is  $w : \mathbb{Z}/2^j\mathbb{Z} \rightarrow \mathbb{R}$  such that for every odd  $n > 0$ ,

$$(1) \quad w(T(n) \bmod 2^j) - w(n \bmod 2^j) < -\log_2 \frac{T(n)}{n}.$$

Consider the family  $n_m = 2^{j+1}m - 1$  for integers  $m \geq 1$ . Since  $j \geq 1$  we have  $n_m \geq 3$ , so every  $n_m$  is a strictly positive odd integer and lies in the domain where (1) is assumed. The accelerated step evaluates exactly to

$$T(n_m) = \frac{3(2^{j+1}m - 1) + 1}{2} = 3 \cdot 2^j m - 1.$$

Reducing modulo  $2^j$ ,

$$n_m = 2^j(2m) - 1 \equiv -1, \quad T(n_m) = 2^j(3m) - 1 \equiv -1 \pmod{2^j}.$$

Substituting  $n_m$  into (1), the potential terms cancel identically, leaving, for every  $m \geq 1$ ,

$$0 < -\log_2 \frac{3 \cdot 2^j m - 1}{2 \cdot 2^j m - 1}.$$

This requires the argument of the logarithm to be strictly less than 1; as both numerator and denominator are positive, that means  $3 \cdot 2^j m - 1 < 2 \cdot 2^j m - 1$ , i.e.  $3 < 2$  — a contradiction.  $\square$

*Remark 3.3* (The mechanism). In  $\mathbb{Z}_2$  the integer  $-1$  is an odd fixed point of the accelerated map:  $T(-1) = (3 \cdot (-1) + 1)/2 = -1$ . Any function of the residues modulo  $2^j$  is blind to the difference between the positive integers  $n_m \equiv -1 \pmod{2^{j+1}}$ , which we want to control, and the point  $-1 \in \mathbb{Z}_2$  they converge to 2-adically, on which the dynamics genuinely gains  $\log_2 \frac{3}{2} = \log_2 3 - 1 > 0$  per step. The family  $n_m$  transports this stationary gain into  $\mathbb{Z}_+$ , where the potential cancels and only the gain remains. The obstruction is thus *structural*: it does not depend on the combinatorics of  $w$ , only on the existence of the 2-adic fixed point.

**Corollary 3.4** (Robustness). *The obstruction of Theorem 3.2 persists under each of the following relaxations of Definition 3.1.*

- (1) Finite exceptions. *If (1) is only required outside a finite set  $E \subset \mathbb{Z}_+$ , no such  $V$  exists.*
- (2) Non-strict decrease. *If the requirement is weakened to  $V(T(n)) \leq V(n)$ , no such  $V$  exists.*

- (3) Blocks of  $k$  iterations. *If the requirement is  $V(T^k(n)) < V(n)$  for a fixed  $k \geq 1$  (for all odd  $n > 0$  whose first  $k$  accelerated steps are odd, or for all odd  $n > 0$  outright), no such  $V$  exists.*
- (4) Restricted domains. *If the decrease is required only on a subset  $D \subseteq \mathbb{Z}_+$  of the odd integers, then either  $D$  omits infinitely many terms of some family  $n_m = 2^{j+1}m - 1$  — in which case  $V$  cannot certify global convergence — or  $D$  contains infinitely many of them, and no such  $V$  exists.*

*Proof.* (1) The test family  $\{n_m\}_{m \geq 1}$  is infinite, so for any finite  $E$  there is  $m$  with  $n_m \notin E$ ; the proof of Theorem 3.2 applies verbatim to that  $m$ .

(2) With  $\leq$  in place of  $<$ , the final inequality of the proof becomes  $3 \cdot 2^j m - 1 \leq 2 \cdot 2^j m - 1$ , i.e.  $3 \leq 2$ , still false.

(3) Parametrize the deeper family  $n_m = 2^{j+k}m - 1$ ,  $m \geq 1$ . We claim  $T^i(n_m) = 3^i 2^{j+k-i}m - 1$  for  $0 \leq i \leq k$ , each of the first  $k$  steps being odd. Indeed, by induction:  $T^0(n_m) = n_m$ ; if  $T^i(n_m) = 3^i 2^{j+k-i}m - 1$  with  $i < k$ , then  $j+k-i \geq j+1 \geq 2$ , so  $3^i 2^{j+k-i}m$  is even and  $T^i(n_m)$  is odd, whence

$$T^{i+1}(n_m) = \frac{3(3^i 2^{j+k-i}m - 1) + 1}{2} = 3^{i+1} 2^{j+k-i-1}m - 1.$$

Hence  $T^k(n_m) = 3^k 2^j m - 1 \equiv -1 \pmod{2^j}$  and  $n_m \equiv -1 \pmod{2^j}$ , so the potentials cancel in the block inequality  $w(T^k(n_m) \bmod 2^j) - w(n_m \bmod 2^j) < -\log_2 \frac{T^k(n_m)}{n_m}$ , leaving

$$0 < -\log_2 \frac{3^k 2^j m - 1}{2^k 2^j m - 1},$$

which forces  $3^k < 2^k$ , false for every  $k \geq 1$ .

(4) If  $D$  contains infinitely many  $n_m$ , run the proof of Theorem 3.2 on those; if it omits all but finitely many, then the orbits through the omitted integers are unconstrained by  $V$ , which therefore proves nothing about them.  $\square$

*Remark 3.5* (Relation to the literature). That *some* obstruction to modular Lyapunov functions must exist over all of  $\mathbb{Z}$  is unsurprising: the negative integers contain genuine cycles ( $-1$ ;  $-5 \rightarrow -7 \rightarrow -10$ ; and the cycle of  $-17$ ; see Section 7.2), and a strictly decreasing  $V$  defined on all of  $\mathbb{Z} \setminus \{0\}$  would forbid them. The content of Theorem 3.2 is sharper: the hypothesis is imposed *only on*  $\mathbb{Z}_+$ , where no cycle other than  $\{1, 2\}$  is known, and the failure is nevertheless unconditional, caused by the projection of the single 2-adic point  $-1$  rather than by any actual cycle in the domain of the hypothesis. We have not located this statement or its proof in the classical or contemporary literature we consulted [10, 6, 2, 5, 8, 7, 9]; we state this as a bibliographic observation, not as a claim of certainty. Section 5.3 recasts the theorem as the infeasibility of a system of difference constraints and Section 7.4 locates the obstruction computationally at every level  $5 \leq j \leq 10$ .

## 4. WASSERSTEIN CONTRACTION OF THE TRANSFER STRUCTURE

This section proves the two contraction theorems. We first fix the probabilistic model, which is standard [9].

**Lemma 4.1** (Geometric law of the 2-adic valuation). *For  $a \geq 1$ , the set of odd integers  $n$  with  $\nu_2(3n+1) = a$  is exactly one residue class modulo  $2^{a+1}$ , consisting of odd residues. Consequently, if  $n$  is drawn uniformly from the odd residues modulo  $2^k$ , then  $\Pr[\nu_2(3n+1) = a] = 2^{-a}$  for every  $1 \leq a \leq k-1$ .*

*Proof.* For odd  $n$ ,  $3n+1$  is even, so  $\nu_2(3n+1) \geq 1$ . For  $a \geq 1$ ,  $\nu_2(3n+1) = a$  iff  $3n+1 \equiv 2^a \pmod{2^{a+1}}$ , i.e.  $n \equiv 3^{-1}(2^a - 1) \pmod{2^{a+1}}$ , a single residue class modulo  $2^{a+1}$  (3 is invertible modulo any power of 2); it consists of odd numbers because  $2^a - 1$  and  $3^{-1}$  are both odd. Among the  $2^{k-1}$  odd residues modulo  $2^k$ , this class contains exactly  $2^{k-a-1}$  of them, giving probability  $2^{k-a-1}/2^{k-1} = 2^{-a}$ .  $\square$

**Definition 4.2** (Syracuse chain on  $\mathbb{Z}_3$ ). Since 2 is a unit in  $\mathbb{Z}_3$ , the maps

$$\varphi_a(x) = (3x+1)2^{-a}, \quad a \geq 1,$$

are well defined on  $\mathbb{Z}_3$ . The *Syracuse chain* is the Markov kernel on  $\mathbb{Z}_3$

$$P(x, \cdot) = \sum_{a \geq 1} 2^{-a} \delta_{\varphi_a(x)}.$$

By Lemma 4.1 this is the transfer model of the Syracuse map under which Tao constructs the Syracuse random variables [9].

**Theorem 4.3** (Contraction on  $\mathbb{Z}_3$ ). *Each branch  $\varphi_a$  scales the 3-adic metric exactly by  $1/3$ :*

$$|\varphi_a(x) - \varphi_a(y)|_3 = \frac{1}{3} |x - y|_3 \quad (x, y \in \mathbb{Z}_3, a \geq 1).$$

*Consequently  $\tau(P) \leq 1/3$ , and the same bound holds for every projection  $P_k$  of the chain onto  $\mathbb{Z}/3^k\mathbb{Z}$  ( $k \geq 1$ ) with the induced 3-adic metric. In particular  $P$  has a unique invariant measure  $\pi$  on  $\mathbb{Z}_3$ ,  $W_1(\mu P^n, \pi) \leq 3^{-n} W_1(\mu, \pi)$  for every  $\mu$ , and  $\text{spec}(U|_{\text{Lip}(\mathbb{Z}_3)}) \subseteq \{1\} \cup \{z : |z| \leq 1/3\}$ .*

*Proof.* For the branch identity:  $\varphi_a(x) - \varphi_a(y) = 3(x-y)2^{-a}$ , and  $|3|_3 = \frac{1}{3}$ ,  $|2^{-a}|_3 = 1$ , so  $|\varphi_a(x) - \varphi_a(y)|_3 = \frac{1}{3} |x - y|_3$  exactly.

For  $\tau(P) \leq 1/3$ , couple  $P(x, \cdot)$  and  $P(y, \cdot)$  *synchronously*: draw a single  $a$  with  $\Pr[a] = 2^{-a}$  and move both points along the same branch  $\varphi_a$ . This is a coupling of the two transition laws, of cost

$$\sum_{a \geq 1} 2^{-a} |\varphi_a(x) - \varphi_a(y)|_3 = \frac{1}{3} |x - y|_3,$$

so  $W_1(P(x, \cdot), P(y, \cdot)) \leq \frac{1}{3} d(x, y)$  for all  $x \neq y$ .

For the projections: on  $\mathbb{Z}/3^k\mathbb{Z}$  put  $d_k(r, s) = 3^{-v_3(r-s)}$  for  $r \neq s$  (well defined since  $v_3(r-s) \leq k-1$ ). The projected kernel  $P_k(r, \cdot)$  is the push-forward of  $P(x, \cdot)$  under reduction modulo  $3^k$ , for any lift  $x$  of  $r$  (the reduction of  $\varphi_a(x)$  modulo  $3^k$  depends only on  $x \bmod 3^k$ , indeed only on  $x \bmod 3^{k-1}$ ). The same synchronous coupling projects to a coupling of  $P_k(r, \cdot)$  and  $P_k(s, \cdot)$ ; each coupled pair  $(\varphi_a(x) \bmod 3^k, \varphi_a(y) \bmod 3^k)$  either coincides (cost 0, which happens when  $v_3(x-y) = k-1$ ) or lies at distance  $3^{-v_3(x-y)-1} = \frac{1}{3}d_k(r, s)$ . Hence  $W_1(P_k(r, \cdot), P_k(s, \cdot)) \leq \frac{1}{3}d_k(r, s)$  and  $\tau(P_k) \leq \frac{1}{3}$ .

The remaining assertions follow from Lemma 2.3 with  $\tau = 1/3$ .  $\square$

*Remark 4.4.* The bound  $\tau(P_k) \leq 1/3$  is uniform in  $k$ , unlike the  $\ell^2$  spectral gap of the finite sections, which is trivial (Section 7.6: the projected chain loses all memory modulo  $3^k$  in exactly  $k$  steps, so all eigenvalues beyond  $\lambda_1 = 1$  vanish). The exact values of  $\tau(P_k)$ , computed in rational arithmetic in Section 7.7, are  $\tau_2 = 5/21$ ,  $\tau_3 = 455/1387$  and  $\tau_4 = 7635497415/22906579627 \approx 0.33333206$ , increasing towards the uniform bound: at finite levels, couplings across distinct branches still help, but their advantage vanishes as  $k \rightarrow \infty$ .

We now turn to the dual chain on  $\mathbb{Z}_2$ . The accelerated map  $T$  extends continuously to  $\mathbb{Z}_2$  and preserves Haar measure; it is conjugate to the one-sided Bernoulli shift [10, 6]. Its transfer (backward) chain is driven by the two inverse branches of  $T$  on  $\mathbb{Z}_2$ :

$$\psi_0(x) = 2x \text{ (even step), } \quad \psi_1(x) = \frac{2x-1}{3} \text{ (odd step),}$$

both well defined on  $\mathbb{Z}_2$  since 3 is a unit in  $\mathbb{Z}_2$ .

**Theorem 4.5** (Contraction and finite-time Haar averaging on  $\mathbb{Z}_2$ ). *Let  $Q$  be the Markov kernel on  $\mathbb{Z}_2$  given by  $Q(x, \cdot) = \frac{1}{2}\delta_{\psi_0(x)} + \frac{1}{2}\delta_{\psi_1(x)}$ , with Koopman operator  $Lf(x) = \frac{1}{2}f(2x) + \frac{1}{2}f(\frac{2x-1}{3})$ . Then:*

- (1)  $\tau(Q) = \frac{1}{2}$  exactly:  $W_1(Q(x, \cdot), Q(y, \cdot)) = \frac{1}{2}|x-y|_2$  for all  $x \neq y$ ;
- (2) Haar measure on  $\mathbb{Z}_2$  is the unique invariant measure of  $Q$ ;
- (3) if  $f : \mathbb{Z}_2 \rightarrow \mathbb{R}$  depends only on  $x \bmod 2^k$  (a level- $k$  cylinder function), then  $L^k f$  is identically equal to the Haar average  $2^{-k} \sum_{r \bmod 2^k} f(r)$ .

*Proof.* Both branches scale the metric by exactly  $1/2$ :  $|\psi_0(x) - \psi_0(y)|_2 = |2|x-y|_2 = \frac{1}{2}|x-y|_2$ , and  $|\psi_1(x) - \psi_1(y)|_2 = |2(x-y)/3|_2 = \frac{1}{2}|x-y|_2$  since  $|3|_2 = 1$ .

(1) The synchronous coupling (same branch on both points) gives cost exactly  $\frac{1}{2}|x-y|_2$ , so  $W_1 \leq \frac{1}{2}|x-y|_2$ . For the reverse inequality, note the cross distances are maximal:  $\psi_0(x) - \psi_1(y) = 2x - \frac{2y-1}{3} = \frac{6x-2y+1}{3}$  has odd numerator and unit denominator, so  $|\psi_0(x) - \psi_1(y)|_2 = 1 \geq \frac{1}{2}|x-y|_2$  (as  $|x-y|_2 \leq 1$ ), and symmetrically. Thus *every* coupling of the two two-point laws transports each unit of mass a distance at least  $\frac{1}{2}|x-y|_2$ , whence  $W_1 \geq \frac{1}{2}|x-y|_2$ .



(2) Uniqueness follows from (1) and Lemma 2.3. For invariance of Haar measure  $\lambda$ :  $\psi_0$  is a homeomorphism of  $\mathbb{Z}_2$  onto  $2\mathbb{Z}_2$  (the even 2-adics) with  $(\psi_0)_*\lambda = 2\lambda|_{2\mathbb{Z}_2}$  (it scales Haar mass by  $|2|_2 = \frac{1}{2}$  onto a set of measure  $\frac{1}{2}$ , uniformly), and  $\psi_1$  is a homeomorphism onto  $1 + 2\mathbb{Z}_2$  (the odd 2-adics:  $2x-1$  is odd and division by the unit 3 preserves 2-adic parity) with  $(\psi_1)_*\lambda = 2\lambda|_{1+2\mathbb{Z}_2}$ . Hence

$$\lambda Q = \frac{1}{2}(\psi_0)_*\lambda + \frac{1}{2}(\psi_1)_*\lambda = \lambda|_{2\mathbb{Z}_2} + \lambda|_{1+2\mathbb{Z}_2} = \lambda.$$

(3) For a word  $u = (u_1, \dots, u_k) \in \{0, 1\}^k$  write  $\psi_u = \psi_{u_1} \circ \dots \circ \psi_{u_k}$ . Then

$$L^k f(x) = 2^{-k} \sum_{u \in \{0,1\}^k} f(\psi_u(x)).$$

Since each branch scales the metric by  $\frac{1}{2}$ ,  $|\psi_u(x) - \psi_u(y)|_2 = 2^{-k}|x - y|_2 \leq 2^{-k}$ , so  $\psi_u(x) \bmod 2^k$  does not depend on  $x$ ; as  $f$  is a level- $k$  cylinder function, each summand is a constant  $c_u$ . We claim the map  $u \mapsto \psi_u(x) \bmod 2^k$  is a bijection onto  $\mathbb{Z}/2^k\mathbb{Z}$ : by induction on  $k$ , the images of  $\psi_0$  and  $\psi_1$  lie in the (disjoint) even and odd classes respectively, and within each class,  $\psi_i$  composed with the  $2^{k-1}$  maps  $\psi_{u'}$  ( $u' \in \{0, 1\}^{k-1}$ ) yields  $2^{k-1}$  points that are pairwise distinct modulo  $2^k$ , because  $\psi_i$  maps points distinct modulo  $2^{k-1}$  to points distinct modulo  $2^k$  (exact  $\frac{1}{2}$ -scaling of the metric). Hence  $\{c_u\}_u$  enumerates  $\{f(r) : r \bmod 2^k\}$  exactly once and  $L^k f \equiv 2^{-k} \sum_r f(r) = \int f d\lambda$ .  $\square$

*Remark 4.6.* Theorem 4.5(3) is the functional-analytic reading of Terras's conjugacy with the Bernoulli shift [10]: the backward chain forgets the level- $k$  observable in exactly  $k$  steps, i.e. mixing is not merely geometric but *finite-time* on cylinder observables. It is also the exact analogue of the finite-level rank collapse of the Syracuse chain (Section 7.6).

*Remark 4.7* (Scope of the contraction theorems). Theorems 4.3 and 4.5 operate on spaces of *global densities* over the compact rings  $\mathbb{Z}_3, \mathbb{Z}_2$ . The set  $\mathbb{Z}_+$  where the conjecture lives is Haar-null in both. Contraction of distributions in  $W_1$  therefore does not, by itself, constrain individual orbits: in  $\ell^1(\mathbb{Z}_+)$ , every finite section of the pushforward operator we tested is nilpotent (spectrum  $\{0\}$ ), and no elementary weight  $n^\theta$  yields uniform contraction — the obstruction being, again, the residue classes  $n \equiv -1 \bmod 2^t$  of Theorem 3.2 (Section 7.7). We state this limitation explicitly to avoid overclaiming: the contraction theorems quantify the measure-theoretic ingredient of “almost-all” results in the style of [10, 9]; they do not approach the pointwise conjecture.

## 5. EXACT ALGORITHMS

This section describes the algorithms behind the verifications of Section 7 and proves the statements they rely on. Implementation details of the arithmetic are deferred to Section 6.

### 5.1. Cycle enumeration via parity vectors.

**Proposition 5.1** (Affine form and cycle equation). *Fix  $d$  odd and let  $p = (p_0, \dots, p_{L-1}) \in \{0, 1\}^L$  be the parity vector of the first  $L$  accelerated steps of  $n$  in the  $3n + d$  system, with  $k = \sum_i p_i$ . Then*

$$T_d^L(n) = \frac{3^k n + b(p)}{2^L},$$

where  $b(p) = b_L$  is given by the recurrence

$$b_0 = 0, \quad b_{i+1} = \begin{cases} 3b_i + d 2^i, & p_i = 1, \\ b_i, & p_i = 0. \end{cases}$$

Consequently  $n$  lies on a cycle of length  $L$  with parity vector  $p$  if and only if

$$(2) \quad n = \frac{b(p)}{2^L - 3^k},$$

which for  $L \geq 1$  has at most one solution ( $2^L \neq 3^k$ ).

*Proof.* By induction on  $i$ : writing  $k_i = p_0 + \dots + p_{i-1}$ , assume  $T_d^i(n) = (3^{k_i} n + b_i)/2^i$  (true for  $i = 0$ ). If  $p_i = 0$ , the next step halves:  $T_d^{i+1}(n) = (3^{k_i} n + b_i)/2^{i+1}$  and  $b_{i+1} = b_i$ . If  $p_i = 1$ ,

$$T_d^{i+1}(n) = \frac{3 T_d^i(n) + d}{2} = \frac{3(3^{k_i} n + b_i)/2^i + d}{2} = \frac{3^{k_i+1} n + 3b_i + d 2^i}{2^{i+1}},$$

matching the recurrence. The cycle condition  $T_d^L(n) = n$  then reads  $n(2^L - 3^k) = b(p)$ .  $\square$

The enumeration algorithm tests (2) for every parity vector: every cycle other than the fixed point 0 contains an odd element, so it may be rotated to begin with  $p_0 = 1$ , and it suffices to sweep the  $2^{L-1}$  vectors with  $p_0 = 1$  for each  $L \leq L_{\max}$ . For each integral solution  $n$ , the algorithm *re-runs the actual orbit* and keeps  $n$  only if its true parity vector is  $p$  and  $T_d^L(n) = n$  — eliminating algebraic solutions that do not realize their parity pattern. The output is therefore the complete list of cycles of  $T_d$  in  $\mathbb{Z}$  of length at most  $L_{\max}$ : exact arithmetic replaces orbit simulation as the search mechanism.

### 5.2. Diophantine cycle exclusion.

**Proposition 5.2** (Cycle inequality). *Suppose every integer  $2 \leq n \leq N$  converges to the trivial cycle under  $T$ , and let  $C$  be a nontrivial cycle of  $T$  in  $\mathbb{Z}_+$  with  $k$  odd elements and total length  $L$  (in  $T$ -steps). Then all elements of  $C$  exceed  $N$  and*

$$(3) \quad 0 < L - k\alpha \leq k \varepsilon(N), \quad \varepsilon(N) := \log_2 \left( 1 + \frac{1}{3N} \right),$$

where  $\alpha = \log_2 3$ . In particular  $\|k\alpha\| \leq k \varepsilon(N)$ .

*Proof.* If  $C$  contained an element  $\leq N$ , that element would converge to the trivial cycle, contradicting nontriviality; so every element exceeds  $N$ . Let  $x_1, \dots, x_k$  be the odd elements of  $C$  in cyclic order, and let  $2^{a_i}$  be the power of two divided out between  $x_i$  and  $x_{i+1}$  (indices mod  $k$ ), so  $x_{i+1} = (3x_i + 1)/2^{a_i}$  and  $\sum_i a_i = L$ . Multiplying around the cycle,

$$2^L = \prod_{i=1}^k \frac{3x_i + 1}{x_i} = \prod_{i=1}^k \left(3 + \frac{1}{x_i}\right).$$

Taking  $\log_2$ :  $L - k\alpha = \sum_i \log_2(1 + \frac{1}{3x_i})$ , which is strictly positive and, since each  $x_i > N$ , at most  $k \log_2(1 + \frac{1}{3N})$ . As  $L$  is an integer,  $\|k\alpha\| \leq L - k\alpha \leq k\varepsilon(N)$ .  $\square$

The exclusion now follows from the classical theory of best rational approximation [4, Ch. I–II]: if  $p_i/q_i$  are the convergents of the (infinite, since  $\alpha$  is irrational) continued fraction of  $\alpha$ , then for  $q_i \leq k < q_{i+1}$ ,

$$\|k\alpha\| \geq \|q_i\alpha\| > \frac{1}{q_i + q_{i+1}},$$

so (3) forces  $k > (\varepsilon(N)(q_i + q_{i+1}))^{-1}$  within that range. Sweeping consecutive convergent denominators yields the largest  $K$  such that every  $k \leq K$  is impossible; then  $L > K\alpha$  bounds the cycle length, and since the  $L$  iterates of a cycle are distinct, also the number of its elements. This is the method of Eliahou [2] and Simons–de Weger [8]; our contribution is the certification scheme (Section 6) making every step exact. Numerical outcomes are in Section 7.3.

**5.3. The obstruction as a graph problem: Karp’s algorithm.** Fix  $j \geq 1$  and let  $M = 2^j$ . Define a weighted digraph  $G_j$  on the vertex set  $\mathbb{Z}/M\mathbb{Z}$ : each residue  $r$  has the two successors  $T(r) \bmod M$  and  $T(r + M) \bmod M$  (the images of the two lifts of  $r$  modulo  $2M$ , which determine  $T$  modulo  $M$ ), and every edge leaving  $r$  carries the weight

$$c(r) = \begin{cases} \log_2 3 - 1, & r \text{ odd,} \\ -1, & r \text{ even.} \end{cases}$$

The weight is the exact asymptotic logarithmic gain of the step on the class: for odd  $n$ ,  $\log_2 \frac{T(n)}{n} = \log_2(\frac{3}{2} + \frac{1}{2n}) \downarrow \log_2 3 - 1$  as  $n \rightarrow \infty$ , and for even  $n$  it is exactly  $-1$ .

**Proposition 5.3** (Necessary feasibility relaxation). *If a modular Lyapunov function of level  $j$  existed (Definition 3.1, extended to require decrease on even steps as well), then the difference constraints*

$$(4) \quad w(s) - w(r) \leq -c(r) \quad \text{for every edge } r \rightarrow s \text{ of } G_j$$

*would be feasible. Conversely, (4) is infeasible if and only if  $G_j$  contains a cycle of strictly positive total weight; and the maximum cycle mean of  $G_j$  equals  $\log_2 3 - 1 > 0$ , attained by the self-loop at the residue  $-1 \bmod M$ .*

*Proof. Necessity.* Let  $r \rightarrow s$  be an edge of  $G_j$  with  $r$  odd, witnessed by the lift class  $\{n \equiv r' \pmod{2M}\}$  (where  $r' \in \{r, r+M\}$ ) whose members map into class  $s$ . For each positive  $n$  in that class, the assumed decrease gives  $w(s) - w(r) < -\log_2 \frac{T(n)}{n} = -\log_2 \left(\frac{3}{2} + \frac{1}{2n}\right)$ . Letting  $n \rightarrow \infty$  along the class, the right side increases to  $-(\log_2 3 - 1) = -c(r)$ , so  $w(s) - w(r) \leq -c(r)$ . For  $r$  even the same argument gives  $w(s) - w(r) \leq 1 = -c(r)$  (with equality of the bound, since  $\log_2 \frac{T(n)}{n} = -1$  exactly).

*Feasibility criterion.* If (4) holds and  $r_0 \rightarrow r_1 \rightarrow \dots \rightarrow r_\ell = r_0$  is a cycle, summing the constraints telescopes to  $0 \leq -\sum_i c(r_i)$ , so every cycle has total weight  $\leq 0$ . Conversely, suppose every cycle of  $G_j$  has total weight  $\leq 0$ , i.e. has cost  $\geq 0$  for the edge costs  $b(r \rightarrow s) := -c(r)$ . Adjoin a super-source  $\sigma$  with a cost-0 edge to every vertex, and set  $w(v)$  equal to the minimum cost of a walk from  $\sigma$  to  $v$ ; this is well defined because walks cannot decrease cost by traversing cycles (all cycle costs are  $\geq 0$ ), so the minimum is attained by a simple path. By minimality,  $w(s) \leq w(r) + b(r \rightarrow s)$  for every edge, which is (4).

*Maximum mean.* Every edge weight is at most  $\log_2 3 - 1$  (as  $\log_2 3 - 1 > -1$ ), so every cycle has mean at most  $\log_2 3 - 1$ . The residue  $r = M - 1 \equiv -1$  is odd, and its lift  $2M - 1$  satisfies

$$T(2M - 1) = \frac{3(2M - 1) + 1}{2} = 3M - 1 \equiv -1 \pmod{M},$$

so  $G_j$  has a self-loop at  $-1 \pmod{M}$  of weight  $\log_2 3 - 1$ , attaining the bound. Its mean is positive precisely because  $3 > 2$ .  $\square$

Proposition 5.3 recasts Theorem 3.2 as the infeasibility of (4) and identifies the culprit cycle a priori. Karp's algorithm [3] computes the maximum cycle mean of  $G_j$  and an attaining cycle by dynamic programming in  $O(|V||E|)$  time; running it for a range of  $j$  (Section 7.4) verifies that the *unique* optimal cycle is the predicted self-loop at  $-1 \pmod{2^j}$  — the computational fingerprint of the 2-adic fixed point.

#### 5.4. Exact transfer kernels and exact $W_1$ on ultrametric quotients.

The projected Syracuse kernel  $P_k$  on  $\{r \pmod{3^k} : 3 \nmid r\}$  is computed exactly: since 2 is a primitive root modulo  $3^k$ , the multiplier  $2^{-a} \pmod{3^k}$  is periodic in  $a$  with period  $\varphi(3^k) = 2 \cdot 3^{k-1}$ , so the total transition mass onto each target residue is a finite geometric aggregate, a closed-form rational. Stationary measures are obtained by Gaussian elimination over  $\mathbb{Q}$ .

Wasserstein distances on the quotient are also exact, via the closed form on ultrametric spaces:

**Lemma 5.4** ( $W_1$  on  $\mathbb{Z}/p^k\mathbb{Z}$ ). *Let  $\mu, \nu$  be probability measures on  $\mathbb{Z}/p^k\mathbb{Z}$  with the metric  $d(x, y) = p^{-v_p(x-y)}$ . For  $1 \leq \ell \leq k$  let  $m_\ell = \frac{1}{2} \sum_{c \pmod{p^\ell}} |\mu(c + p^\ell\mathbb{Z}) - \nu(c + p^\ell\mathbb{Z})|$  be the total-variation discrepancy of the projections modulo*

$p^\ell$ . Then

$$W_1(\mu, \nu) = \sum_{\ell=1}^{k-1} (p^{-(\ell-1)} - p^{-\ell}) m_\ell + p^{-(k-1)} m_k.$$

*Proof.* For  $x \neq y$  the ultrametric distance decomposes along the tree of congruence cells: writing  $[x \not\equiv y (p^\ell)]$  for the indicator,

$$d(x, y) = \sum_{\ell=1}^{k-1} (p^{-(\ell-1)} - p^{-\ell}) [x \not\equiv y (p^\ell)] + p^{-(k-1)} [x \neq y],$$

since the sum telescopes to  $p^{-v_p(x-y)}$ . Integrating against any coupling  $\gamma$  and using that the mass  $\gamma\{x \not\equiv y (p^\ell)\}$  is at least the total-variation distance  $m_\ell$  of the two projections modulo  $p^\ell$  gives the lower bound  $W_1 \geq \sum_\ell c_\ell m_\ell$  with the stated coefficients  $c_\ell \geq 0$ .

For the upper bound, construct a coupling greedily along the cell tree, from the finest level to the coarsest. First match, within each cell  $c$  modulo  $p^k$ , the common mass  $\min\{\mu(c), \nu(c)\}$  at distance 0. The unmatched mass at level  $k$  has total  $m_k$  on each side; match as much of it as possible within cells modulo  $p^{k-1}$  (each such match travels distance  $p^{-(k-1)}$ ), leaving unmatched totals  $m_{k-1}$  on each side — indeed, after all mass is matched within level- $(k-1)$  cells wherever possible, the surplus of  $\mu$  over  $\nu$  remaining in the cell  $c'$  modulo  $p^{k-1}$  is  $(\mu(c') - \nu(c'))^+$ , whose total is exactly  $m_{k-1}$ . Continuing upward, at each level  $\ell = k-1, \dots, 1$  the mass newly matched within level- $\ell$  cells is  $m_{\ell+1} - m_\ell$ , at distance  $p^{-\ell}$  (mass conservation makes the within-cell surpluses independent of how the finer levels were matched), and finally  $m_1$  is matched across level-1 cells at distance 1. The total cost is

$$\sum_{\ell=1}^k p^{-(\ell-1)} (m_\ell - m_{\ell-1}) \quad (m_0 := 0),$$

which by Abel summation equals  $\sum_{\ell=1}^{k-1} (p^{-(\ell-1)} - p^{-\ell}) m_\ell + p^{-(k-1)} m_k$ , matching the lower bound.  $\square$

The contraction coefficient  $\tau(P_k) = \max_{x \neq y} W_1(P_k(x, \cdot), P_k(y, \cdot)) / d(x, y)$  is then a maximum of finitely many exact rationals. The values reported in Remark 4.4 were obtained this way.

## 6. METHODOLOGY: EXACT ARITHMETIC

Because part of our evidence is computational, we specify the arithmetic model precisely; referees of computer-assisted results rightly demand this. The guiding rule is: *no floating-point quantity participates in any certified claim*. Concretely:

- (1) **Integers and rationals.** All integer arithmetic uses Python's built-in arbitrary-precision integers (no overflow is possible; numbers such as  $3^q$  with  $q \approx 10^5$  are handled exactly). All rational arithmetic (transfer matrices, stationary measures, Wasserstein distances, the

coefficients  $\tau_k$ , the cycle equation (2)) uses exact fractions from the standard `fractions` module. Reported values such as  $\tau_3 = 455/1387$  are exact outputs, with decimal expansions given only as display approximations.

- (2) **The constant  $\log_2 3$ .** The only irrational constant in the toolkit enters at two places, each handled without trusting floating point:
  - *Convergents* (Section 5.2). The partial quotients of  $\alpha = \log_2 3$  are extracted from a 120-digit decimal approximation (`decimal` module, correctly rounded to the working precision), and every convergent  $p/q$  with  $q \leq 10^5$  is then *certified exactly*: the predicted side of the alternation  $p/q \geq \alpha$  is equivalent to the integer comparison  $2^p \geq 3^q$ , which is checked in exact integer arithmetic. The certification failing would raise an error; it never does within the range used by our bounds.
  - *The slack  $\varepsilon(N)$*  (Proposition 5.2). The algorithm replaces  $\varepsilon(N)$  by a certified rational *upper* bound (the 120-digit value rounded up by one unit in the last place). Any overestimate of  $\varepsilon$  only weakens the resulting bound  $K$ , never its validity.
- (3) **Where floats do appear.** Karp’s algorithm (Section 5.3) runs its dynamic program on IEEE double weights for speed. Its output is not trusted as such: the value of the maximum cycle mean is *proved* to be  $\log_2 3 - 1$  (Proposition 5.3), and the run is accepted only if the extracted optimal cycle consists solely of odd residues — a symbolic condition checked exactly, under which the cycle’s mean is  $\log_2 3 - 1$  by construction. The float value agrees with  $\log_2 3 - 1$  to  $< 2 \cdot 10^{-15}$  in all runs, as expected. Likewise, the power-iteration estimate of the second eigenvalue of  $P_k$  is a numerical diagnostic only; the corresponding certified statement is the exact rank collapse  $P_k^k = \mathbf{1}\pi_k$  verified in rational arithmetic (Section 7.6).
- (4) **Determinism.** All algorithms are deterministic; there is no Monte Carlo component anywhere in the toolkit. Re-running the reproduction script yields bit-identical results on any platform (the test suite runs on CPython 3.9–3.13 on Linux and macOS in continuous integration).

## 7. COMPUTATIONAL VERIFICATION

Every number in this section is regenerated and checked by the reproduction script `reproduce_paper_results.py`; the labels R1–R8 refer to its sections. We emphasize (cf. Section 1.2) that items 7.1–7.5 are *verifications with explicit finite limits*, not theorems, while the exact rational outputs in 7.6–7.7 are finite-level instances of Theorems 4.3 and 4.5.

**7.1. Convergence sieve (R1).** Every  $1 \leq n \leq 2 \cdot 10^5$  reaches the trivial cycle under  $T$  (ascending sieve: each  $n$  is iterated only until it drops below itself). This is far below the published record  $2^{68}$  [1] and is included solely as

a self-contained, re-runnable hypothesis for 7.3. The largest stopping time in the range (accelerated steps until the orbit first drops below its seed) is attained at  $n = 35,655$ , with 135 steps; the largest excursion is attained at  $n = 159,487$ , whose orbit peaks at 8,601,188,876.

**7.2. Cycle enumeration (R2).** The enumeration of Section 5.1, swept over all parity vectors of length  $\leq 14$ :

| System                       | Cycles found (by least- $ \cdot $ element)                               | Status                |
|------------------------------|--------------------------------------------------------------------------|-----------------------|
| $3n + 1$ , in $\mathbb{Z}$   | $\{1, 2\}; \{-1\}; \{-5, -7, -10\}$ ; cycle of $-17$ (len. 11)           | complete, $L \leq 14$ |
| $3n - 1$ , in $\mathbb{Z}_+$ | $\{1\}; \{5, 7, 10\}$ ; cycle of 17 (len. 11) ( $\dagger$ )              | validation            |
| $3n + 5$ , in $\mathbb{Z}_+$ | $\{1, 4, 2\}; \{5, 10\}; \{19, 31, 49, 76, 38\}; \{23, 37, 58, 29, 46\}$ | validation            |

( $\dagger$ ) For  $3n - 1$  the listed cycles are the positive ones; by the conjugation  $n \mapsto -n$  they are exactly the negative cycles of  $3n + 1$  (verified exactly:  $\varphi(x) = -x$  satisfies  $\varphi \circ T_1 = T_{-1} \circ \varphi$ , R8).

The enumeration over  $\mathbb{Z}$  is *complete* up to the swept length: on the positives of  $3n + 1$ , only the trivial cycle exists with  $L \leq 14$ ; the three negative cycles are rediscovered by the algorithm, validating the detector on known ground truth.

**7.3. Diophantine exclusion (R3).** Proposition 5.2 with the certified convergents of  $\alpha$ :

| Hypothesis (all $n \leq N$ converge)   | odd steps $k$ of any cycle | elements            |
|----------------------------------------|----------------------------|---------------------|
| $N = 2 \cdot 10^5$ (verified here, R1) | $k > 428$                  | $> 676$             |
| $N = 2^{71}$ (cf. [1])                 | $k > 65,470,613,320$       | $> 103,443,569,045$ |

The second row uses the verification limit reported by the ongoing project of [1]; with the published  $2^{68}$  the bound is of the same order. These reproduce the method of [2, 8].

**7.4. Locating the obstruction (R4).** Karp's algorithm on  $G_j$  for  $j = 5, \dots, 10$ : in every run the maximum mean equals  $\log_2 3 - 1 = 0.5849625\dots$  (float agreement  $< 2 \cdot 10^{-15}$ ; value certified symbolically per Section 6), and the optimal cycle returned is *unique and equal to the self-loop at  $-1 \bmod 2^j$*  predicted by Proposition 5.3 — the finite-level fingerprint of Theorem 3.2.

**7.5. Inverse tree of the trivial cycle (R5).** The conjecture is equivalent to the statement that the preimage tree of 1 under  $T$  (edges  $m \rightarrow 2m$  and, when integral and odd,  $m \rightarrow (2m - 1)/3$ ) covers  $\mathbb{Z}_+$ . Exact BFS gives: the minimum depth at which the tree rooted at 1 covers all of  $\{1, \dots, 1000\}$  is 113 (the depth of  $n = 871$ , equivalently the maximum total stopping time in the range); and at level 30 the new nodes span exactly the interval of values  $[123, 2^{31}]$  (minimum and maximum new node). Growth per level is  $\approx 4/3$ , consistent with the branching heuristic; Krasikov–Lagarias prove the reached set below  $X$  has cardinality  $\geq X^{0.84}$  [5].

**7.6. Finite-level Syracuse spectra (R6).** For the projected kernel  $P_k$  (exact rationals, Section 5.4): the uniform measure is *not* stationary; the exact stationary measure modulo 3 is  $\pi(1) = 1/3$ ,  $\pi(2) = 2/3$  — Syracuse iterates fall on  $2 \bmod 3$  twice as often as on  $1 \bmod 3$ , an exact consequence of  $(3n+1)/2^a \equiv (-1)^a \pmod{3}$  with  $\Pr[a \text{ odd}] = 2/3$  under Lemma 4.1. Moreover  $P_k^k = \mathbf{1}\pi_k$  *exactly* (rank collapse verified in rational arithmetic for the tested  $k$ ): the chain loses all memory modulo  $3^k$  in exactly  $k$  steps, so the  $\ell^2$  spectrum beyond  $\lambda_1 = 1$  is  $\{0\}$  — which is why the surviving contraction structure must be sought in the Wasserstein geometry (Theorem 4.3), not in finite  $\ell^2$  sections.

**7.7. Exact Wasserstein coefficients (R7).** Lemma 5.4 applied to the exact kernels:

| Level                                 | exact $\tau_k$             | decimal           |
|---------------------------------------|----------------------------|-------------------|
| $3^2$                                 | $5/21$                     | $0.238095\dots$   |
| $3^3$                                 | $455/1387$                 | $0.328046\dots$   |
| $3^4$                                 | $7635497415/22906579627$   | $0.33333206\dots$ |
| $\mathbb{Z}_3$ (Theorem 4.3)          | $\leq 1/3$ (uniform bound) | $0.\overline{3}$  |
| $2^6$ on $\mathbb{Z}_2$ (Theorem 4.5) | $1/2$ (exact)              | $0.5$             |

The coefficients increase monotonically towards the uniform bound  $1/3$ , and each branch-contraction identity, the projective coherence of the family  $\pi_k$ , and the finite-time Haar averaging  $L^k f = \int f d\lambda$  of Theorem 4.5(3) are verified in exact arithmetic at the tested levels. On the other side of Remark 4.7, the finite sections of the pushforward operator on  $[1, N] \subset \mathbb{Z}_+$  are exactly nilpotent (acyclic transition structure) for the tested  $N \leq 10^5$ , and the weight  $n^\theta$  fails to contract uniformly on the classes  $n \equiv -1 \bmod 2^t$ , as predicted by Theorem 3.2.

**7.8. Structural cross-checks (R8).** Terras’s bijection (parity vector bijective on  $\mathbb{Z}/2^{14}\mathbb{Z}$ ) verified exactly; the affine conjugation  $\varphi(x) = -x$  between  $3n+1$  and  $3n-1$ , and the embedding  $\varphi(x) = 5x$  of  $3n+1$  into  $3n+5$ , verified exactly on the tested ranges.

## 8. DISCUSSION

**What is proved.** Theorem 3.2 and Corollary 3.4 close off, unconditionally, a family of elementary strategies: no potential of finitely many residue classes modulo a power of two can convert the average logarithmic descent of  $T$  into a monotone certificate on  $\mathbb{Z}_+$ , and the failure survives every standard relaxation. The cause is a single 2-adic point,  $-1$ , invisible to modular arithmetic yet arbitrarily well approximated by positive integers. Any successful Lyapunov-type argument must therefore break the 2-adic symmetry between  $\mathbb{Z}_+$  and the negatives — for instance by using unbounded corrections or non-periodic information. Theorems 4.3 and 4.5 identify the norm



in which the transfer structure of the map does contract uniformly (Wasserstein over the compact  $p$ -adic rings, coefficients  $1/3$  and  $1/2$ ), in contrast with the trivial  $\ell^2$  spectra of all finite sections.

**What is verified.** The computations of Section 7 validate the implementations on systems with known nontrivial cycles ( $3n - 1$ ,  $3n + 5$ , and the negatives of  $3n + 1$ ), instantiate Theorem 3.2 at the levels  $j = 5, \dots, 10$ , and exhibit the exact finite-level coefficients  $\tau_k$  ascending to the bound of Theorem 4.3. None of these computations is offered as evidence of the conjecture itself, and our sweep limits are modest by design: the heavy verification frontier belongs to [1].

**Limits and open directions.** Remark 4.7 delimits what contraction of global densities can give:  $\mathbb{Z}_+$  is Haar-null in  $\mathbb{Z}_2$  and  $\mathbb{Z}_3$ , so  $W_1$ -contraction does not constrain individual orbits, and our own finite-section experiments show the pointwise operator on  $\mathbb{Z}_+$  is spectrally trivial. The quantified frontier left open by this toolkit is concrete: cycle lengths below  $10^{11}$  elements are excluded (Section 7.3); divergence requires an orbit forever avoiding a strictly negative mean drift with binomial fluctuations [10, 9]; and any modular-potential proof strategy is excluded outright (Theorem 3.2). What remains, in either direction, must live beyond finite residue arithmetic.

#### DATA AND CODE AVAILABILITY

The Python package `collatz`, the test suite, the reproduction script `reproduce_paper_results.py`, and this manuscript are available in the accompanying repository under the MIT license (code) and CC BY 4.0 (text). All results reproduce with CPython  $\geq 3.9$  with no third-party dependencies.

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